

CSE 460
VLSI DESIGN SECTION-02

ASSIGNMENT-2

- 1) Derive the Moore type state diagram, state table, state assigned table and the final circuit for a modulo-6 up counter (i.e. a counter that counts 0,1,2,3,4,5,0,1,2,...).
 - It should have a 1-bit input w , a clock signal and a multi-bit output z .
 - The machine should increase its count whenever $w = 1$, and hold its previous count otherwise.
 - At each counting step the machine should output the value of the count in binary.

- 2) Design both a moore and mealy machine that detects the following pattern in its input (w) : 1, 0, 1. Draw the state diagram, state table, state assigned table and the final circuit.
 - Repeating and overlapping sequences should be detected.
 - The machine should generate the output $z = 1$ if in the past 3 clock cycles 101 pattern was detected, and $z = 0$ otherwise.

- 3) An FSM that has an input w and an output z . The machine has to generate $z = 1$ when the previous four values of w were 1001 or 1111; otherwise, $z = 0$. Overlapping input patterns are allowed. An example of the desired behavior is:

w : 010111100110011111
 z : 000000100100010011

4)

Consider a sequential circuit that detects if two consecutive bits are the same or not. The circuit generates an output $z=1$ if two consecutive bits are opposite and $z=0$ if the bits are the same. The output changes in the same clock cycle where the input is received.

w	0	0	1	0	0	1	1	0
z	0	0	1	1	0	1	0	1

Consider overlapping bit sequence

- (a) **Determine** the type of FSM required for solving this problem. 1
- (b) **Draw** the state diagram of the circuit. 3
- (c) **Derive** a state-assigned table from (b) using the gray-encoding technique. 3
- (d) **Find** the next-state and output logic expressions using k-map. 3

5)

Consider a sequential circuit with two inputs ($p1$ & $p2$) and a single output, z . The decisions regarding the status of various product orders are handled by two departments: the Sales Team and the Warehouse. Both departments independently approve (denoted by 1) or reject (denoted by 0) an order. The decisions of the Sales Team are given by a bit stream, $p1$, while the decisions of the Warehouse are represented by $p2$. If, for two consecutive orders, both departments approve (i.e., $p1=p2=1$ for two consecutive clock cycles), the sequential circuit provides $z=1$ at the next order. The decisions of the Sales Team & the Warehouse for several orders and their corresponding synchronization instances are given below:

p1	0	1	1	1	1	1	1	0	0	0	1
p2	1	1	1	1	1	0	1	0	0	1	1
z	0	0	0	1	0	1	0	0	0	0	0

Note:

- You must use binary encoding to assign states.
- The overlapping of the input bits ($p1$ & $p2$) is not allowed.

- (a) **Determine** the type of FSM that would be needed for the given problem? 1
- (b) **Design** the state diagram for the corresponding FSM 2
- (c) **Derive** the state table and state-assigned table from your state diagram in (b). 3
- (d) **Evaluate** the next state & output logic expressions to implement the FSM & calculate the number of flip-flops required for the circuit implementation. (You don't need to draw the circuit implementation). 4